Canary Deployment Using Lambda and API Gateway

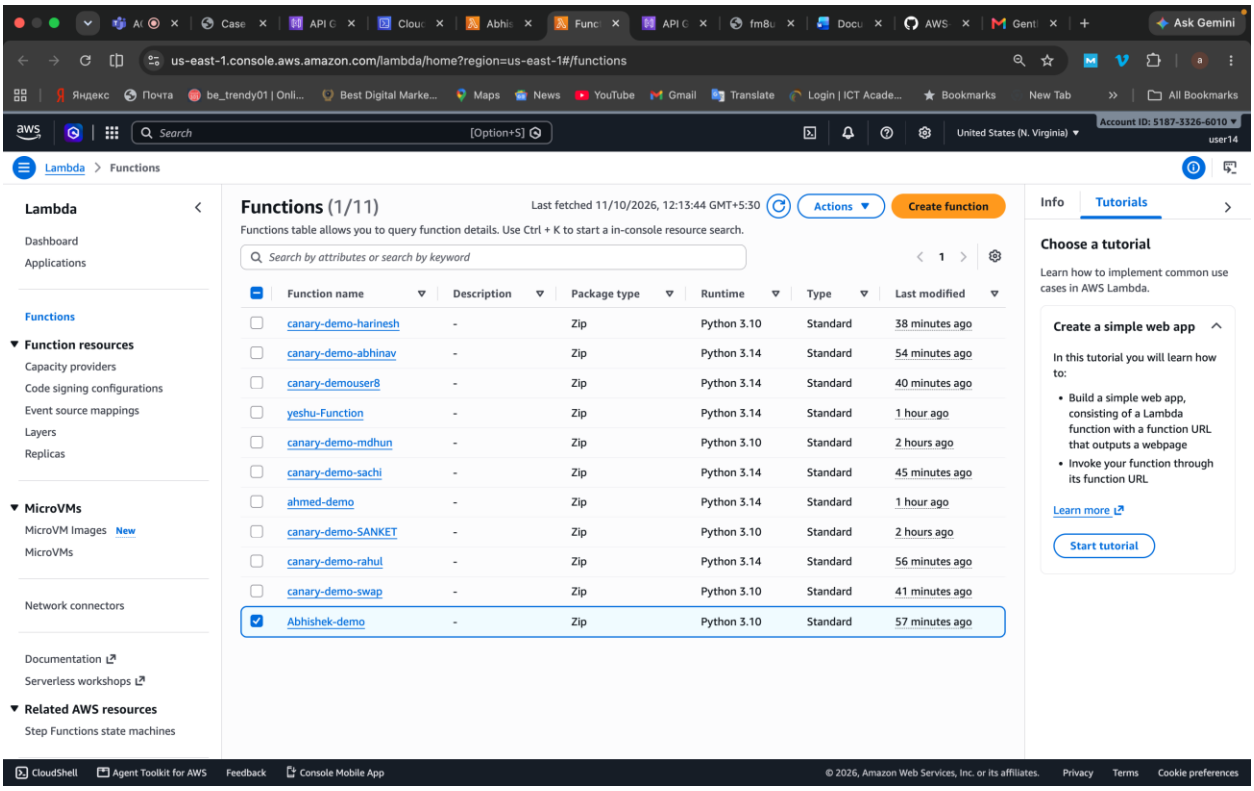
Problem Statement:

The ABC Company is continuously updating the application software. The application is deployed on Lambda and API Gateway. Recently new version of the application is being released. The CTO of the company wants to check the new version of the application with a sample set of traffic and if everything is working fine then all100% of traffic will be redirected. As application developer you are instructed to use native feature of API Gateway for doing so.

Solution

Task 1: Create a Lambda function

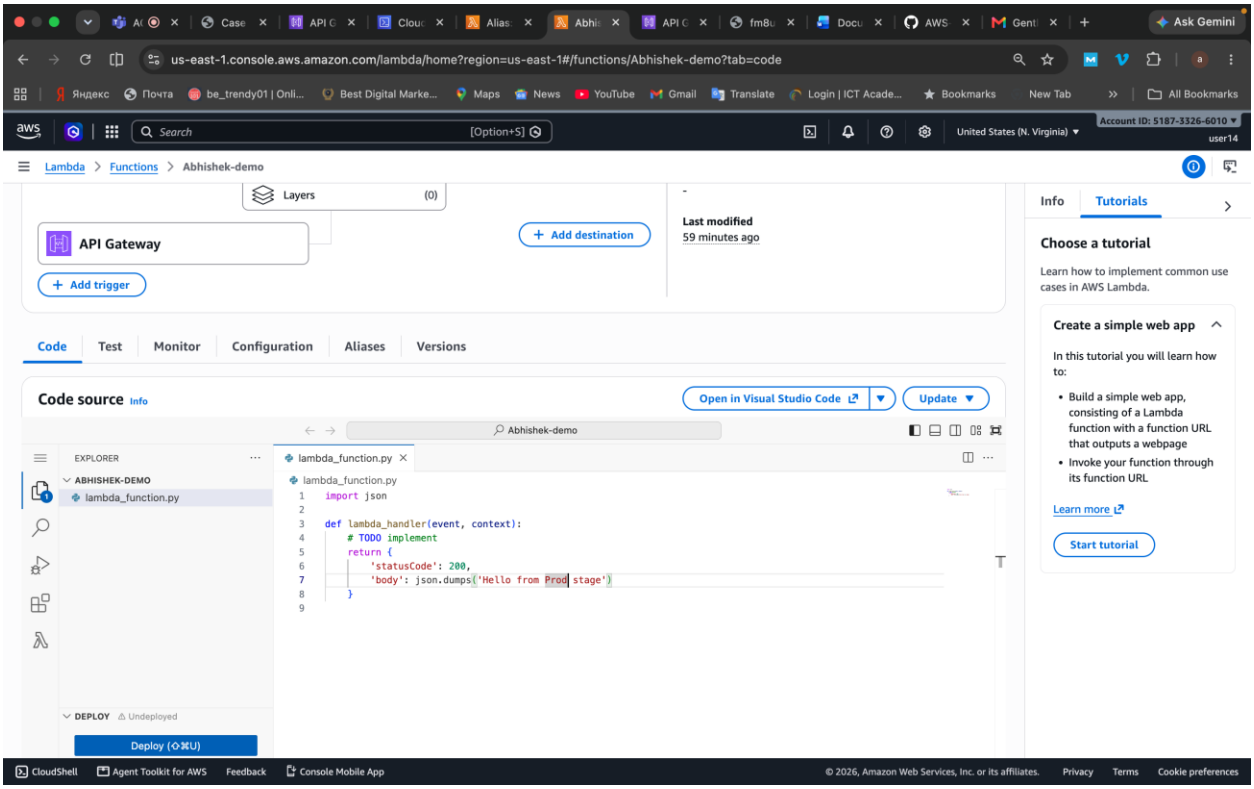
1. Go to Lambda Console
2. Choose Functions from Navigation Pane.
3. Select Create Function
4. Choose Author from Scratch
5. Enter canary-demo as function Name
6. For runtime, select Python 3.10 or latest Python Runtime



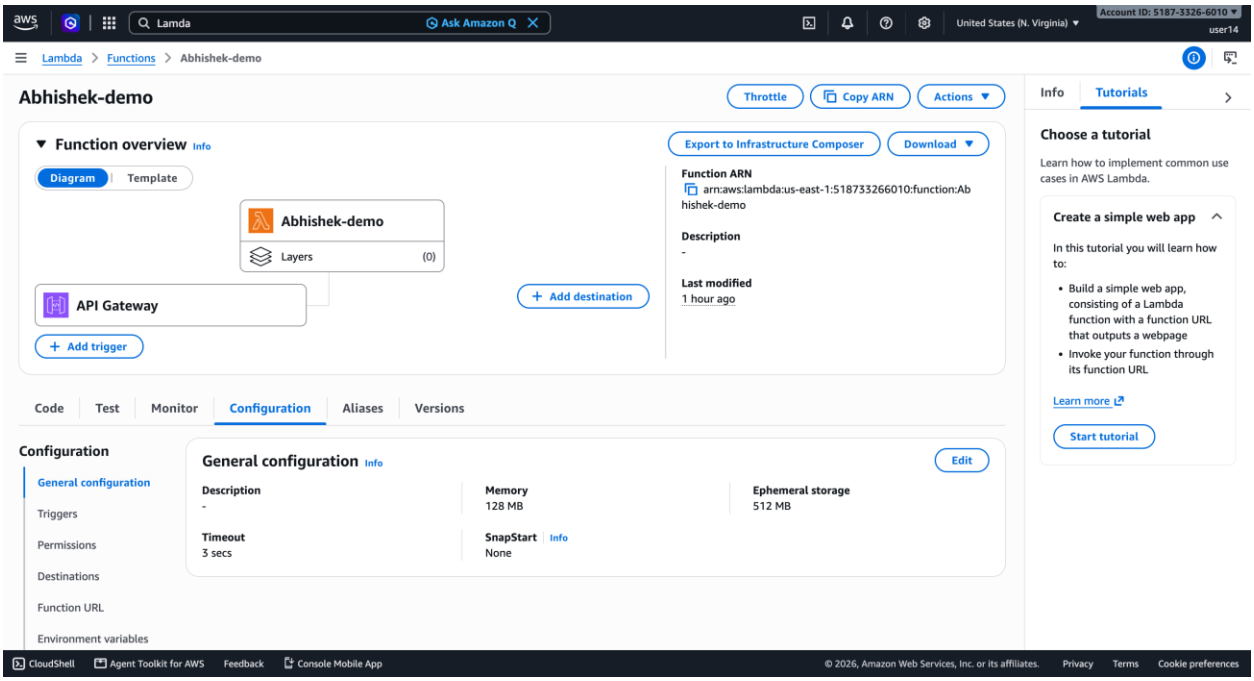
7. Select Create function.

8. Once the function is created, change the return text “Hello from prod stage” instead of “Hello from Lambda”

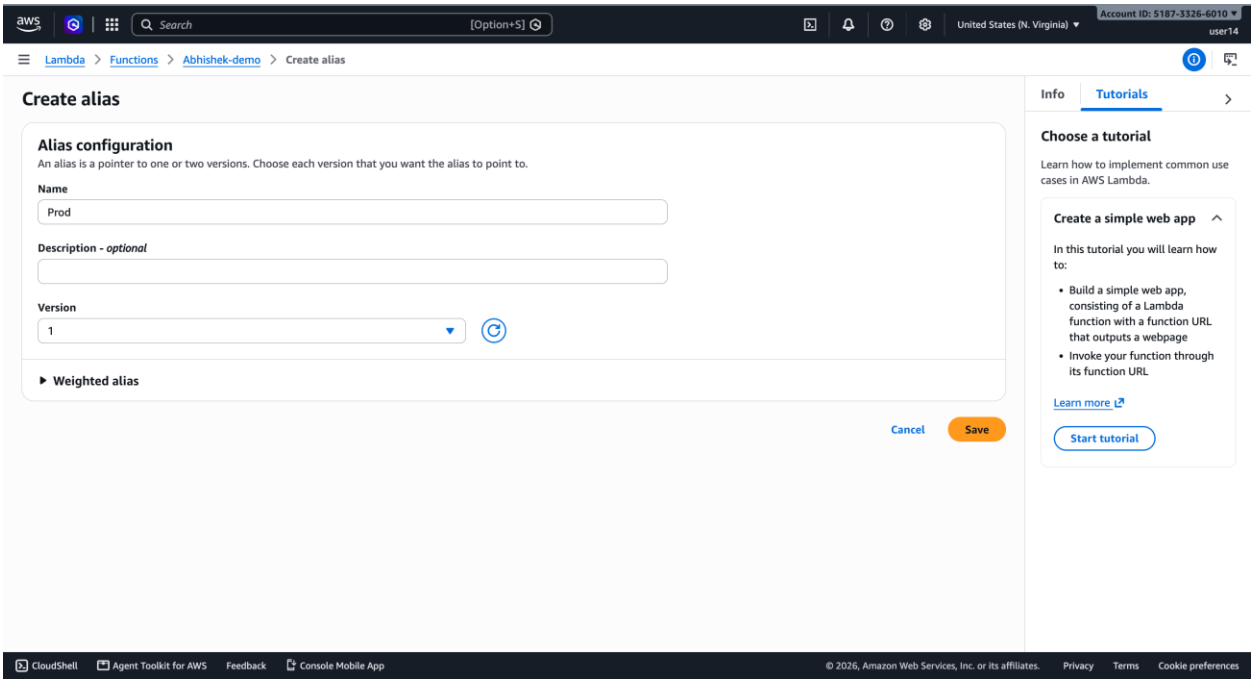
9. Deploy the changes.



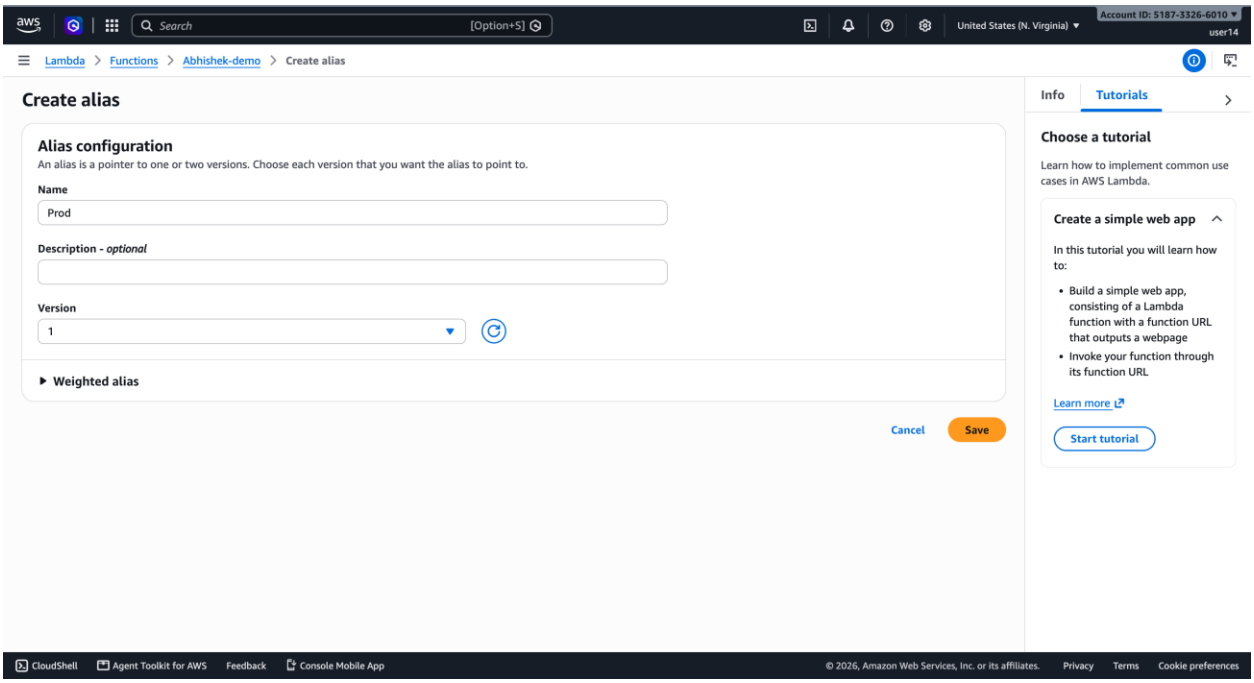
10. Go to action and select publish new version



1. Put a description as “prod” and press publish



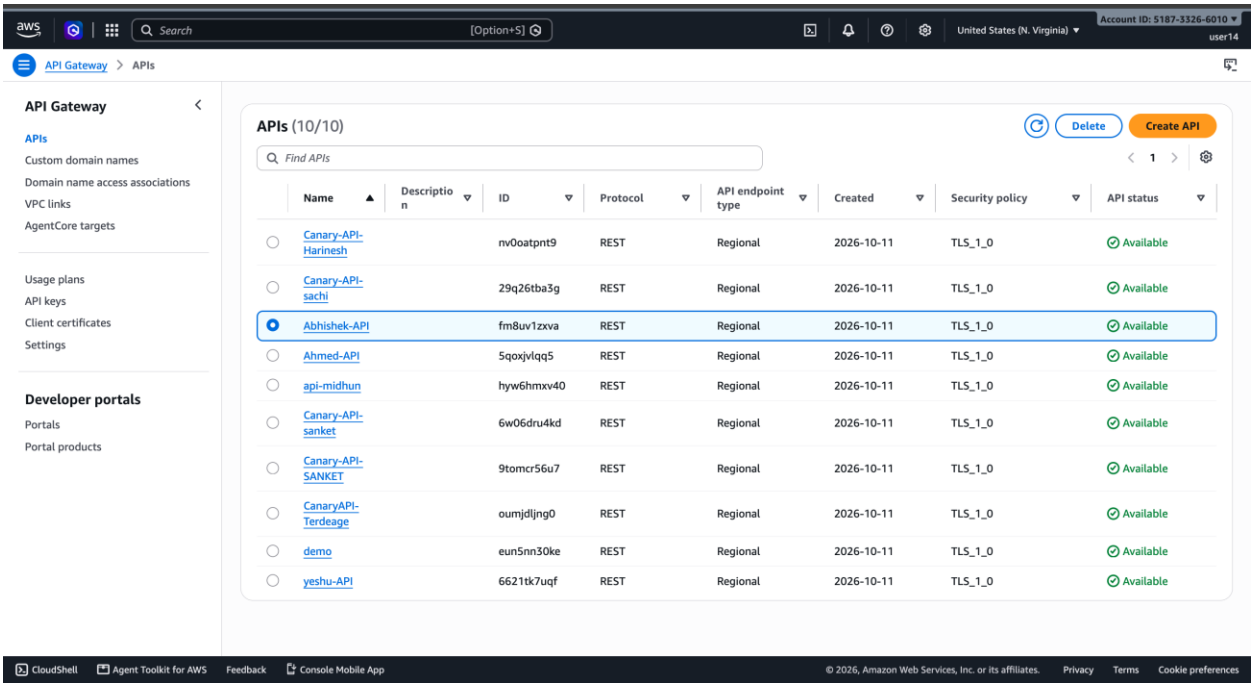
12. Go to action again, select Create alias



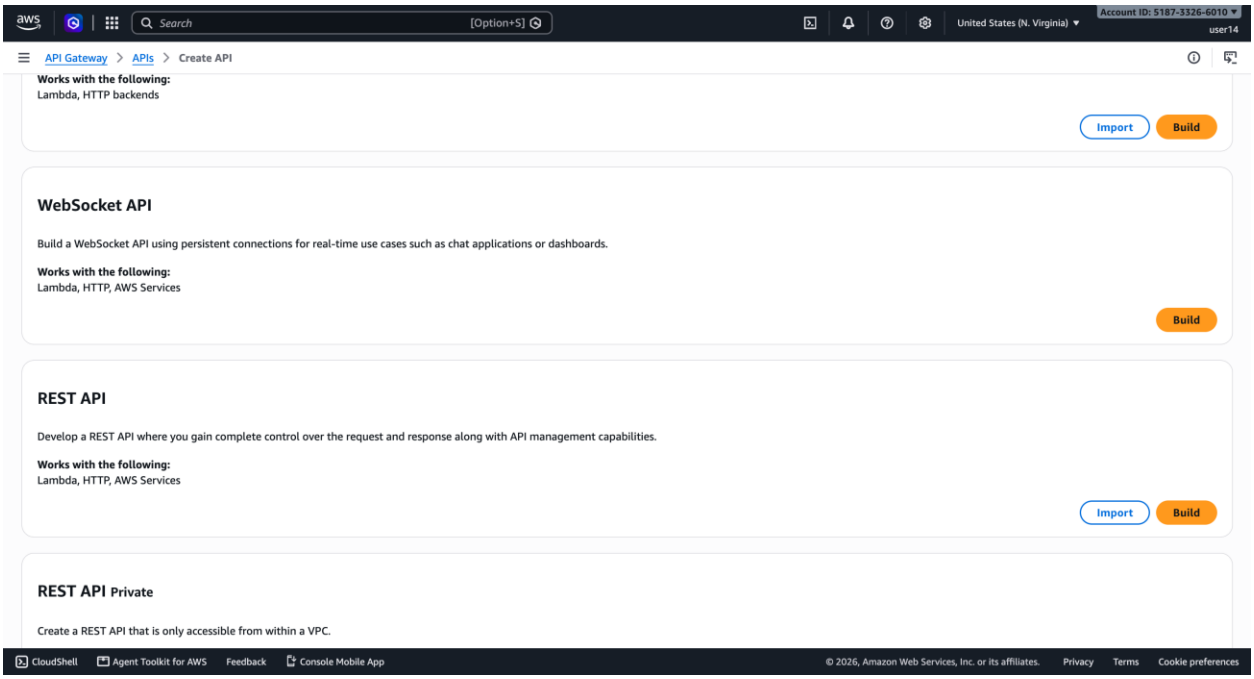
Task 2: Create an API Gateway

1.Go to API Gateway Console and select Create API

2. Under create API window select Rest based API and press Build button



2. Under create API window select Rest based API and press Build button



3. Enter name as Canary-API

4.Under Resource name , enter resource name as get data

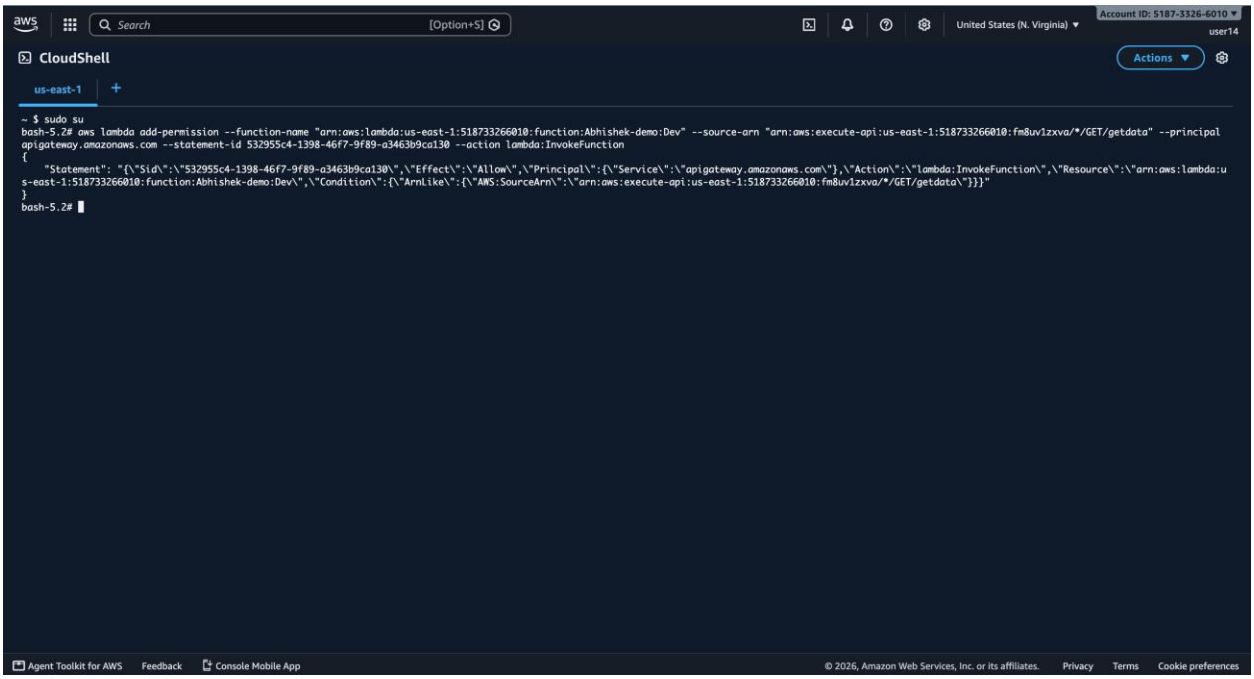
6. Go to action again under resource , select create method and select GET from list

9. In Lambda Function,Enter name of Lambda function followed by stage variable name as follows- canary-demo:${stageVariables.version} where version is name of stage variable and press save.

11. Once you press Save button it will ask you to add permissions

12. Copy and run those commands using AWS CLI. 13. Replace name of alias with stage variable in command as follows

aws lambda add-permission --function-name "arn:aws:lambda:us-east-1:518733266010:function:Abhishek-demo:Prod" --source-arn "arn:aws:execute-api:us-east-1:518733266010:fm8uv1zxva/*/GET/getdata" --principal apigateway.amazonaws.com --statement-id 532955c4-1398-46f7-9f89-a3463b9ca130 --action lambda:InvokeFunction


Search
[Option+S]
United States (N. Virginia)
Account ID: 5187-3326-6010
user14
CloudShell
Actions
us-east-1
~ $ sudo su
bash-5.2# aws lambda add-permission --function-name "arn:aws:lambda:us-east-1:518733266010:function:Abhishek-demo:Dev" --source-arn "arn:aws:execute-api:us-east-1:518733266010:fm8uv1zxva/*/GET/getdata" --principal apigateway.amazonaws.com --statement-id 532955c4-1398-46f7-9f89-a3463b9ca130 --action lambda:InvokeFunction
{
"Statement": "{\"Sid\":\"532955c4-1398-46f7-9f89-a3463b9ca130\",\"Effect\":\"Allow\",\"Principal\":{\"Service\":\"apigateway.amazonaws.com\"},\"Action\":\"lambda:InvokeFunction\",\"Resource\":\"arn:aws:lambda:us-east-1:518733266010:function:Abhishek-demo:Dev\",\"Condition\":{\"ArnLike\":{\"AWS:SourceArn\":\"arn:aws:execute-api:us-east-1:518733266010:fm8uv1zxva/*/GET/getdata\"}}}"
}
bash-5.2#
Agent Toolkit for AWS
Feedback
Console Mobile App
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